An 8-Dimension Playtest Rubric as a Research Instrument for AI-Simulated Narrative Game Sessions

Gio Della-Libera
Independent Research
Seattle, WA, USA
giodl@microsoft.com

Abstract

Measuring the quality of AI-simulated tabletop role-playing game sessions is an unsolved problem: when no human players participate, there are no exit surveys, no post-session interviews, and no participant accounts of what was experienced. We present the Marathon Playtest Rubric, an 8-dimension instrument covering Engagement, Mechanical Fairness, Pacing, Character Agency, Module Fidelity, Atmospheric Landing, Surprise, and Table Readiness, each scored 0–10 for a maximum of 80 points per session. We validate the rubric across 49 AI-simulated Dungeons & Dragons 5th Edition sessions spanning 7 complete campaigns with distinct party archetypes and central dramatic questions. Mean gate scores improved from 65/80 at campaign inception (rubric v1.0, sessions 1–2) to 77/80 at campaign maturity (v1.12, sessions 43–49), a 12-point improvement driven by 13 incremental amendments. The three most recent campaigns score within 1.4 points of the four earliest campaigns, demonstrating that quality measurements are consistent across campaign archetype and theme. The rubric's amendment pipeline — innovations logged, clustered, and codified as new anchors — is itself characterized as a learning mechanism applicable beyond TRPG research.

Keywords

playtest rubric, AI dungeon master, TRPG simulation, narrative games, session quality measurement, amendment-driven evaluation

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1 Introduction

Tabletop role-playing games (TRPGs) are collaborative narrative systems in which a Dungeon Master (DM) designs and adjudicates a fictional world while players embody characters who interact with that world [28]. The quality of a TRPG session is a compound phenomenon: it depends simultaneously on mechanical fairness (did the dice produce outcomes that felt earned?), narrative momentum (did scenes move at the right pace?), character engagement (did each player-character contribute meaningfully?), emotional landing (did designed payoffs register as genuinely affecting?), and design fidelity (did the adventure deliver what it promised?). Human participants navigate all of these dimensions implicitly during play, and their retrospective judgments — whether a session was "good" — are shaped by all of them at once [5].

Measuring this compound quality systematically has long been recognized as difficult. Most professional game development relies on qualitative exit surveys, post-session interviews, and practitioner heuristics rather than structured, multi-dimensional scoring instruments [12, 16]. Academic treatments of playtest methodology focus primarily on deficit detection — the process of identifying where the game breaks — rather than on measuring the degree to which a session succeeded across a defined quality space [30].

The rise of large language model (LLM) systems capable of functioning as autonomous DMs introduces a new version of this measurement problem that is both more tractable and more urgent. When an AI system simultaneously plays the DM and all player characters, the session is fully logged, dice rolls are reproducible, and every narrative decision is documented. But human participants are absent: there are no exit surveys to administer, no participant accounts to aggregate, and no experiential reports to rely on. The only record of session quality is the session log itself — the structured narrative produced by the AI pipeline. Evaluating that log requires a scoring instrument that can operate systematically on textual evidence alone.

This is the problem the Marathon Playtest Rubric was built to solve. The rubric is an 8-dimension scoring instrument, each dimension scored 0–10 via evidence-anchored band descriptions, applied at the close of every session to produce a gate score out of 80. It was developed and validated within the Marathon D&D Workshop, a structured research environment for designing and playtesting Dungeons & Dragons 5e adventures set in the Dragonlance campaign setting. All sessions were run by an AI system (Claude, Anthropic) functioning as DM, with all player characters also simulated by that system from detailed character-sheet specifications. The result is a corpus of 49 fully logged, fully reproducible AI-simulated sessions across 7 complete campaigns.

1.1 The Measurement Problem for AI-Run TRPG

Existing quality frameworks for TRPGs assume human participants and rely on self-report [5, 24]. Player-type models such as Bartle's taxonomy [4] characterize player preferences but do not provide

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session-level scoring. Immersion and affect frameworks [6, 20] require subjective reports of experiential states that simulated player-characters cannot provide. Narrative coherence metrics from AI story generation research [2, 11, 23] evaluate whether generated text is well-formed and plausible, but a session can be narratively coherent and still fail as a TRPG experience: if player-characters never make meaningful choices, if designed emotional payoffs do not land, or if dice mechanics and story systematically undermine each other, the session has failed by TRPG standards regardless of how well-formed the prose is.

Text evaluation metrics from natural language processing — BLEU [22], ROUGE [19], BERTScore [31] — measure similarity to a reference rather than the presence of specific quality properties. LLM-as-judge approaches [32] are promising but introduce the circularity of evaluating an AI system using the same class of system that produced the output. Property-based evaluation — checking that outputs contain evidence of specified outcomes — avoids this circularity and does not require a gold-standard corpus [7]. The Marathon rubric adopts this approach: every score requires specific textual evidence from the session log; general impressions are explicitly prohibited by the scoring protocol.

1.2 Why Session-Level Scoring Matters for TRPG Research

Session-level scoring enables analyses that aggregate per-session observations do not. By scoring each of 49 sessions on the same 8-dimensional instrument, we can track quality trajectories across campaigns, compare opening sessions to closing sessions, and compare campaigns with fundamentally different party archetypes on the same scale. When the rubric is also an adaptive instrument — amended when empirical use reveals gaps — the amendment history is itself a finding: it maps the quality space of AI-simulated TRPG more precisely than any *a priori* specification could.

Beyond TRPG specifically, the Marathon rubric offers a template for any research context in which AI-generated creative outputs need systematic evaluation at scale without participant feedback. The combination of property-based anchors, dimensional independence, and forward-only amendment is applicable to AI-generated fiction, AI-facilitated collaborative design, and AI-simulated social scenarios more broadly.

1.3 Contributions

This paper makes four contributions to the study of AI-simulated narrative game sessions:

(1) **The 8-dimension rubric specification.** We describe the complete rubric with per-dimension anchor language, band boundaries (0–3, 4–6, 7–9, 10), and scoring protocol, including cross-dimensional interaction anchors that emerged from empirical use. The rubric is described at sufficient detail to be adopted or adapted by other researchers.
(2) **A 49-session, 7-campaign validation corpus.** We report gate scores across all 49 sessions of the Marathon corpus, including per-dimension means, binding PASS rates, and the gate/panel spread metric that diagnoses structural artifacts in individual sessions. This is, to our knowledge, the largest published corpus of scored AI-simulated TRPG sessions.
(3) **An amendment-driven evolution trace.** We document all 13 rubric versions from v1.0 to v1.12, identifying the specific empirical events (session innovations) that triggered each amendment. This trace demonstrates how a property-based scoring rubric self-calibrates toward a maturing creative corpus.
(4) **Cross-campaign consistency evidence.** We show that gate scores are consistent across campaigns with fundamentally different archetypes (grief-themed, professional-code, duty-spine, party-of-strangers), confirming that the rubric measures session execution quality rather than thematic content.
(5) **Construct validity analysis.** We discuss construct validity through panel agreement analysis and scope remaining validity gaps requiring human inter-rater studies.

1.4 Paper Organization

Section 2 surveys related work in playtest methodology, AI as dungeon master, quality measurement in creative and narrative systems, and expert review simulation. Section 3 describes the Marathon workshop pipeline in full, the 8-dimension rubric with all anchor language, the amendment mechanism, and the seven campaigns. Section 4 presents the empirical results: gate score trajectories, per-dimension trends, version-controlled opening scores, cross-campaign comparisons, and binding-threshold PASS rates. Section 5 analyzes what the 12-point improvement measures, discusses cross-campaign consistency, and enumerates limitations and implications. Section 6 summarizes contributions and identifies priorities for future work.

2 Related Work

2.1 Playtest Methodology in Game Design

Playtesting is the primary quality-assurance mechanism in game development, yet it has received limited systematic treatment in the academic literature. Practitioner accounts dominate: Fullerton's *Game Design Workshop* [12] articulates playtesting as iterative discovery of design failures, with evaluators focused on moments when the game "breaks" rather than on assessing the degree to which designed qualities are present. This deficit-oriented model treats playtesting as a debugging process. Hunicke, LeBlanc, and Zubek's MDA framework [16] provides a vocabulary for distinguishing mechanics, dynamics, and aesthetics in game analysis, but does not provide a scoring instrument.

Zagal et al. [30] study collaborative games and identify dimensions of cooperative quality (shared decision-making, role complementarity, joint problem solving) that prefigure several of the Marathon rubric's concerns. Their analysis is category-level rather than scored, and addresses board games rather than improvisational narrative games. Deterding et al. [9] characterize game design elements in terms of their contribution to motivational quality, which aligns with the Character Agency dimension but does not provide a measurement framework.

The Marathon rubric differs from prior playtest instruments in three ways. First, it is both deficit-oriented (penalizing failures in Mechanical Fairness, Module Fidelity) and surplus-oriented (rewarding the presence of emergent positive outcomes in Surprise

and Atmospheric Landing). Second, it produces a numerical score rather than a categorical judgment, enabling trajectory analysis across sessions. Third, it is explicitly designed to operate on session logs rather than participant reports, making it applicable to AI-simulated sessions where no human participants are present.

2.2 AI as Dungeon Master

Automated narrative systems have a long research history, from interactive drama frameworks [23] to procedural quest generation [10] to character-driven story engines [2]. The primary evaluation challenge across this literature is that quality metrics designed for static narrative (coherence, plausibility, narrative arc completeness) do not capture the improvisational and participatory qualities specific to role-playing contexts.

Akoury et al. [1] introduce STORIUM, a dataset and evaluation platform for machine-in-the-loop story generation that shares the Marathon corpus's concern for structured narrative quality evaluation. STORIUM uses human ratings and card-based constraint systems; it does not address the fully-automated setting where no human participants are present. Fan, Lewis, and Dauphin [11] study strategies for LLM story generation with a focus on narrative structure but evaluate through human raters reading story outputs, a method that does not scale to large session corpora.

The AI Dungeon system [27] demonstrates that large language models can produce coherent narrative in TRPG-like contexts at scale, but its evaluation relies on player engagement signals (session continuation, sharing behavior) rather than structural session quality assessment. Mirowski et al. [21] examine co-writing scripts and theatrical scenes with AI, identifying quality dimensions (character voice consistency, dramatic arc coherence, staging feasibility) that partially overlap with the Marathon rubric's Character Agency and Module Fidelity dimensions. Their evaluation uses theatrical practitioners as raters rather than an automated scoring instrument.

2.3 Quality Measurement in Creative and Narrative Systems

The broader literature on quality measurement for AI-generated text identifies three primary approaches: reference-based metrics, preference-based metrics, and property-based metrics [7]. Reference-based metrics (BLEU [22], ROUGE [19]) measure similarity to a gold-standard corpus; they require a reference corpus and cannot evaluate properties that have no single correct expression. Preference-based metrics (pairwise comparison, rating scales) collect human judgments but do not scale to large corpora without crowdsourcing infrastructure [1].

Property-based metrics check that outputs satisfy specified properties, independently of any reference. Andrade [3] develops scoring rubrics in educational assessment contexts that are explicitly property-based: each rubric criterion defines a property that a successful work should exhibit, and the rubric score reflects the degree to which evidence of that property is present. This is the model the Marathon rubric follows.

LLM-as-judge approaches [32] have gained prominence as a scalable alternative to human rater studies, using a language model to evaluate outputs from a language model. Liang et al. [18] apply this approach to research paper review with encouraging results for factual dimensions. The Marathon rubric explicitly avoids this approach for its primary scoring, opting for rule-based evidence citation (a specific passage from the log must be cited for every non-zero sub-score) rather than model judgment. However, the five-lens panel review that runs alongside the rubric gate does use AI-as-expert simulation, following the approach described by Gero and Kan [13] for design critique and extended by Mirowski et al. [21] for creative work evaluation.

Yuan et al. [29] study collaborative story writing with LLMs using a tool that offers completions, continuations, and elaborations to human authors. Their evaluation focuses on the human author's experience (satisfaction, perceived quality, creative ownership), which is orthogonal to the Marathon rubric's focus on automated session log quality. Lee, Liang, and Yang [17] similarly study human-AI co-writing and find that the dimensions humans care about (agency, expressiveness, surprise) are multi-dimensional and do not reduce to a single quality axis, a finding consistent with the Marathon rubric's 8-dimensional structure.

2.4 Expert Review Simulation

Simulated expert review — using an AI system to adopt the perspective of a domain expert and evaluate creative or technical work — has been studied in design critique [13], literary and dramatic evaluation [21], and research peer review [18]. The common finding is that structured role assignment (giving the AI a specific expert identity and evaluation rubric) significantly improves the specificity and consistency of the produced evaluation compared to open-ended critique requests.

The Marathon workshop uses two expert-simulation panels operating at different points in the session pipeline. A five-persona design review panel (Gygax, Jaquays, Schick, Moldvay, Patrick Stuart) evaluates compiled adventure modules before play, scoring on axes specific to each designer's documented aesthetic priorities. A five-lens player-experience panel (Tactician, Roleplayer, Lorekeeper, Improviser, Fresh Face) evaluates session logs after play from five distinct player perspectives. These panels are distinct from the rubric-based session gate and serve complementary rather than redundant functions: the gate scores evidence of outcomes in the session log; the panels evaluate whether those outcomes would satisfy specific types of players or design critics.

Tychsen et al. [26] study how player type affects TRPG experience across platforms, identifying that the same session can succeed for one player type and fail for another. This finding motivates the Marathon rubric's approach of using multiple player-experience lenses in the panel rather than a single aggregate player judgment.

3 Methodology

3.1 The Marathon Workshop Pipeline

The Marathon D&D Workshop is a structured pipeline for designing and playtesting D&D 5e adventures set in the Dragonlance campaign setting. Every session follows an identical seven-stage sequence that produces a reproducible, fully logged session artifact from a compiled adventure module and a character-sheet-specified party. The stages are:

(1) **PREP.** The adventure is compiled to a module document that inlines all room descriptions, encounter stat blocks, treasure

manifests, and critical rule text. The party's character sheets are loaded. The applicable rubric version is recorded in the session frontmatter. Dice seeds are initialized.

(2) **PLAY.** An AI system (Claude, Anthropic) runs the session as DM while simultaneously playing all player characters according to their Playstyle Heuristics — per-PC behavioral specifications that govern when each character speaks, acts tactically, pursues personal goals, and defers to other party members. All dice rolls use a seeded pseudorandom number generator for reproducibility.

(3) **LOG.** The session log is produced as a structured narrative document recording all scenes, dice rolls with natural values and applied modifiers, PC actions and narration, and tagged SURPRISE moments where outcomes exceeded design prediction.

(4) **GATE.** The session log is scored against the locked rubric version using the session-gate protocol described in Section 3.4, producing a per-dimension scorecard and an overall gate score out of 80.

(5) **PANEL.** Five player-experience lenses (Tactician, Roleplayer, Lorekeeper, Improviser, Fresh Face) each review the session log from their distinct perspective and produce a weighted aggregate panel score. Gate and panel scoring are performed independently.

(6) **INNOVATE.** The session log and panel reviews are scanned for outcomes that existing rubric anchors cannot adequately score. Qualifying outcomes are logged as innovations with a dimension tag. When two or more logged innovations cluster in the same dimension, an amendment proposal is drafted per the protocol in Section 3.3.

(7) **HANDOFF.** A canonical handoff document is produced that records all campaign-permanent facts established in the session, updates the TRACKER with session metrics, and registers any new seeds or retrieved seeds in the campaign seed log. No session is considered complete without a handoff.

3.2 The 8-Dimension Rubric

The rubric scores each session on eight dimensions, each scored 0–10 using four band levels: 0–3 (significant failure), 4–6 (adequate but limited), 7–9 (strong evidence of presence), and 10 (full anchor requirements met). The maximum session score is 80. Dimensions are designed to be independent: a session's score on one dimension should not mechanically constrain its score on another, except where explicit cross-dimensional anchors have been ratified (see MF below).

Engagement (Eng). Did each player character contribute meaningfully to the session's outcomes? A score of 0–3 indicates that one or more PCs were passive observers for most of the session, with no scene they initiated or driven by their character-specific motivation. A score of 4–6 indicates that all PCs were present and acted but that their contributions were primarily reactive rather than character-motivated. A score of 7–9 requires that every PC drove at least one scene and that each PC's contribution is traceable to something specific on their character sheet. A score of 10 requires the additional condition that the PCs' combined contributions produced an outcome that the adventure's module did not pre-script — a genuinely emergent collective result.

Mechanical Fairness (MF).. Did dice rolls produce outcomes that felt honestly earned, and did mechanics serve the fiction? The 0–3 band covers sessions where dice results were narratively arbitrary (high roll produces low-value outcome, or vice versa) or where a mechanic functioned in a way that undermined player choice. The 4–6 band covers sessions where dice and outcomes were consistent but mechanics were applied narrowly, without narrative integration. The 7–9 band requires that every significant dice result is integrated into the narrative without retconning and that at least one mechanic is used in a way that extends a character's established fictional positioning (e.g., an Athletics roll that both succeeds mechanically and advances a character's established physical competence in-fiction). The 10-band requires, in addition to the 7–9 criteria, a *cross-character mechanical compound*: at least one instance where one PC's class ability or spell is applied to another PC's roll, framed as an in-character action rather than metagame resource management. This anchor was added at v1.4 after Session 6 of Campaign 1 produced the first instance of this phenomenon (a Divination Wizard invoking her Portent die on behalf of a named ally while speaking in her mentor's voice).

A separate MF anchor, ratified at v1.1, caps MF at 6 when any mechanic in the session systematically damages outcomes in other dimensions. This *cross-dimensional-harmlessness* requirement reflects the finding from Session 1 that a Constitution-save exhaustion mechanic caused two atmospheric payoffs to fail through mechanical side-effect rather than narrative logic, producing false Atmospheric Landing failures that were not captured by any initial anchor.

Pacing (Pac). Did scenes earn their length, and did the session's arc feel appropriately shaped? The 0–3 band covers sessions where one or more scenes ran significantly past or short of their narrative content — a combat encounter that extended for three rounds after the decisive action was resolved, or a revelation scene cut so short that its emotional content had no time to register. The 4–6 band covers sessions with adequate pacing in most scenes but one notable structural mismatch. The 7–9 band requires that every scene's length is proportional to its narrative weight, including transition scenes and downtime scenes. The 10-band adds the requirement that the session's overall arc is visible in the session log: a reader with no prior knowledge of the campaign should be able to identify a beginning, a middle, and an end to the session as a self-contained narrative unit.

Character Agency (CA).. Did player characters make choices that the module required them to make, and did those choices shape the outcome? The 0–3 band covers sessions where outcomes were determined by module logic regardless of what PCs chose. The 4–6 band covers sessions where PCs chose from a menu of options but the options were all equivalent in narrative consequence. The 7–9 band requires that at least one session outcome depended specifically on a PC choice that the module anticipated but did not require — the module offered the choice; the PC took it; and taking it led to a materially different outcome than not taking it would have produced. The 10-band was amended twice (v1.9, v1.10) to add

two additional qualifying paths: a *sustained behavioral pattern* (a sequence of non-actions across multiple scenes that collectively constitutes the choice, with each individual non-action ambiguous but the pattern deliberate) and a *character-register conversion* (a PC converts a mechanical situation into a character-authored outcome traceable to their character sheet heuristics rather than tactical calculation, such as ending a combat encounter by naming an NPC's coercion rather than dealing damage).

Module Fidelity (MFid). Did the adventure deliver the outcomes and payoffs it was designed to deliver? The 0–3 band covers sessions where one or more designed payoffs failed entirely — the artifact was not found, the NPC arc did not fire, the climactic scene was skipped. The 4–6 band covers sessions where designed payoffs occurred but in reduced form. The 7–9 band requires that all designed anchor symptoms were present in the session. The 10-band adds the requirement that at least one hidden payoff — an outcome the module designed but did not make explicit in the anchor list — was also delivered. Amendment v1.2 added a *passive-Perception fallback*: if a primary anchor symptom fails because no PC rolled high enough to trigger it, a passive-Perception delivery of the same information (via environmental description rather than active investigation) counts as a valid anchor-symptom delivery. Amendment v1.3 added a *positional-redundancy* requirement: at the 7–9 band and above, anchor symptoms must have two of three fallback delivery paths available in the session log, ensuring that module design provides sufficient redundancy rather than placing all delivery weight on a single roll.

Atmospheric Landing (AL).. Did designed emotional and sensory moments land as affecting? Atmospheric Landing received the most amendments of any dimension (v1.1, v1.2, v1.4, v1.5, v1.6, v1.7, v1.8, v1.9, v1.12) because novel positive outcomes most frequently exceeded its initial anchors. The 0–3 band covers sessions where designed atmospheric moments are present in the log but are not integrated into PC responses — the description is there but no PC acknowledgment, silence, or behavioral change follows. The 4–6 band covers sessions where at least one atmospheric moment is acknowledged by a PC but the acknowledgment is brief and does not persist into subsequent scenes. The 7–9 band requires one moment that shows evidence of genuine impact: either a specific sensory or emotional image that is subsequently referenced (verbally or behaviorally) by at least one PC, or a moment of sustained silence or non-response that the session log itself flags as significant.

The 10-band evolved through successive amendments to encompass six distinct qualifying paths, any one of which suffices: (a) a specific image that will outlive the session, acknowledged by at least one PC through an inter-PC chain (v1.1); (b) a multi-scene silent-reception chain where a PC holds an unarticulated understanding across three or more scenes and acts on it without naming it (v1.4); (c) an act-without-announcement where a PC acts differently as a direct consequence of a prior naming event, without declaring the connection (v1.5); (d) simultaneous arc-convergence where a single naming act triggers multiple outstanding arc-completions in distinct PCs (v1.6); (e) a framework-failure arc-activation where a supporting NPC's expected role collapses and a PC steps into the resulting gap without prompting (v1.7); and (f) an intra-session restraint anchor where a PC demonstrates the same act-without-announcement behavior within the same session (rather than cross-session), signaling a behavioral change still in the moment of formation (v1.12).

Surprise (Sur). Did anything genuinely emergent happen that the adventure design could not have predicted? The Surprise dimension explicitly rewards outcomes that were not designed. The 0–3 band covers sessions with no logged surprise moments — every outcome was fully predictable from the module and the character sheets. The 4–6 band covers sessions with one or two minor surprises (unexpected dice results, minor improvised NPC responses). The 7–9 band requires at least one substantive emergent outcome that changed the adventure's trajectory or produced a narrative development the module did not anticipate. The 10-band requires that this outcome is not only substantive but amendment-worthy: it represents a quality that existing rubric anchors cannot capture, or it demonstrates that a party archetype produces session dynamics not previously seen in the corpus.

Table Readiness (TR).. Could a DM pick up this session's compiled module and run it without external references? This dimension evaluates the module and session infrastructure rather than the session narrative. The 0–3 band covers sessions where key rule text was absent from the compiled module (requiring lookup during play), NPCs lacked stat blocks, or encounter balance was unvalidated. The 4–6 band covers sessions where critical references were inline but supplementary rules required lookup. The 7–9 band requires that all rule text necessary to run every encounter, trap, and hazard in the session is inlined in the module document, NPCs have complete stat blocks, and encounter balance has been validated against party tier and size. The 10-band adds the requirement that the module's DM cheatsheet includes the three most likely player-decision branches and their adjudication, enabling the DM to handle the session's principal choice points without improvisation under pressure.

3.3 The Amendment Pipeline

The rubric's amendment mechanism is the instrument through which empirical use refines the quality space it can measure. The pipeline has five steps:

(1) **Innovation identification.** At the INNOVATE stage of the session pipeline, the session log and panel reviews are read with the question: did any session outcome exceed the predictive power of the current rubric anchors? An outcome that is clearly captured (positive or negative) by existing anchor language does not qualify. An outcome that required the reviewer to reason outside the rubric's anchor set qualifies as an innovation candidate.
(2) **Innovation logging.** Qualifying outcomes are logged to the innovations register with a dimension tag (identifying which rubric dimension is most relevant), a scope classification (anchor-gap: the dimension's existing anchors cannot capture the outcome; or new-dimension-candidate: the outcome suggests a quality orthogonal to all eight existing dimensions), and a verbatim log citation.
(3) **Clustering and proposal.** When two or more logged innovations carry the same dimension tag, an amendment

proposal is drafted. The proposal describes the pattern in generalized terms (not specific to the source session), specifies the band to which the new anchor applies, and justifies why the pattern represents a genuinely new form of the dimension's quality rather than a restatement of existing anchors.

(4) **Ratification.** The proposal is reviewed against three criteria: the pattern is genuinely novel (not already captured by existing anchor language through reasonable interpretation); the pattern is reproducible (it could in principle appear in future sessions with different parties and adventures); and the proposed anchor language is precise enough to enable consistent scoring. If all three criteria are met, the proposal is adopted, the rubric version is incremented, and the constituent innovations are marked as adopted in the register.

(5) **Forward-only application.** The new anchor applies to all sessions scored after the amendment date. Sessions scored under prior versions are not retroactively rescored. Cross-version comparisons are made explicit by reporting the rubric version alongside gate scores.

This forward-only constraint preserves the integrity of the historical score series: each session score reflects the quality space that was operationally defined at the time it was played. Retrospective revision would conflate the rubric's learning with genuine quality change, making it impossible to distinguish a 12-point improvement in session quality from a 12-point expansion of the rubric's scoring ceiling.

3.4 Session-Gate Scoring Protocol

For each dimension, the session-gate reviewer reads the session log with that dimension as sole focus. The reviewer identifies supporting evidence (specific quoted passages from the log) and counter-evidence (absences, failures, or degraded outcomes), assigns a score using the band descriptions, and writes an evidence cell that documents the specific log citations on which the score rests. The scoring protocol explicitly prohibits generous or harsh adjustment: "Do not score generously out of sympathy. Do not score harshly to seem rigorous. Score what is in the log."

The binding pass threshold is 56/80 (70%), set conservatively to be achievable by a well-designed session scoring 7/10 on all eight dimensions without requiring near-perfect performance. Sessions 1–3 of each campaign are scored as advisory only, because adventures during this window are still under active revision and the session may be the first time the module has been run. Advisory scores contribute to the innovation log but do not determine pass/fail. Sessions 4 onward are binding.

3.5 The Seven Campaigns

The corpus consists of seven complete campaigns, each comprising seven adventures and seven sessions, for a total of 49 sessions. Campaigns were designed with distinct party archetypes and central dramatic questions to test whether the rubric's scores depend on thematic content or reflect session execution quality independent of theme.

Campaign 1 (Moon-Silver Cycle) used a grief-themed party with explicit loss-hooks on every character sheet, centered on a campaign question of what an emotion is worth trading away. Campaign 2 (Conclave Compact) used a covenant-restoration party structure with a convergence deadline generating mechanical time pressure. Campaign 3 (Halted Spire) used a faith-and-authority theme with a party defined by a crisis of institutional legitimacy. Campaign 4 (Thorngate Watch) introduced resource exhaustion mechanics (Supply Die, no long rests for seven sessions) with a siege-and-resource management archetype. Campaign 5 (Thieves Guild) used a professional-code motivation with no grief hooks and no designed personal stakes, testing whether the rubric's Atmospheric Landing dimension depends on grief-adjacent party design. Campaign 6 (Military Unit) used a duty-spine with an institutional deliverable. Campaign 7 (Party of Strangers) used a party with no designed personal stakes, no grief paragraphs, no loyalty objects, and no professional codes — the most stripped-down archetype in the corpus, designed to test whether the rubric can score sessions when parties have no designed emotional infrastructure.

4 Evaluation

4.1 Study Overview

The evaluation corpus consists of 49 sessions across 7 complete campaigns, run between April 18 and April 22, 2026. All sessions were scored using the session-gate protocol described in Section 3.4, with the applicable locked rubric version recorded in each session's frontmatter. Sessions 1–3 of each campaign are advisory (21 sessions total); sessions 4–7 are binding (28 sessions total). All 28 binding sessions exceeded the 56/80 pass threshold (100% PASS rate).

4.2 Gate Score Trajectory

Table 1 reports gate scores for all 49 sessions organized by campaign and session position. The trajectory shows a consistent within-campaign rising pattern from advisory opening sessions to binding mature sessions, with the most pronounced improvement between session 1 (advisory) and session 4 (first binding session).

The minimum gate score in the corpus is 64 (C3-S1, opening advisory session under rubric v1.6). The maximum is 79 (C2-S7, campaign finale under rubric v1.6). The mean across all 49 sessions is 74.0 (SD 3.6). The mean across binding sessions only is 75.2 (SD 2.1). The restricted spread in binding sessions (SD 2.1 versus 3.6 for all sessions) reflects the advisory window absorbing much of the within-campaign variance — early sessions under active design revision drive the lower tail.

4.3 Campaign-Opening Score Controlled for Rubric Version

A primary analysis concern is whether the corpus-wide gate score improvement from C1 (65/80 mean) to later campaigns reflects genuine quality improvement in session design and execution or is an artifact of rubric amendment making the same quality easier to score. Table 2 addresses this by comparing campaign-opening session scores (session 1 of each campaign) under the applicable rubric version at that time.

Table 1: Gate Score Trajectory: All 49 Sessions by Campaign and Session Position. Advisory sessions (S1–S3) are shown in brackets. Rubric version applicable to each campaign's binding sessions is noted. C1–C4 mean: 73.7; C5–C7 mean: 74.0.

Camp.	Party	Version	S1	S2	S3	S4	S5	S6	S7	Mean	SD
C1 Moon-Silver	varduin-muster	v1.0–v1.4	[65]	[65]	[74]	76	78	78	78	74.1	5.2
C2 Conc. Compact	compact-wardens	v1.5–v1.6	[72]	[77]	[78]	76	77	75	79	76.3	2.5
C3 Halted Spire	the-witnesses	v1.6–v1.8	[64]	[70]	[70]	72	74	75	77	71.7	4.4
C4 Thorn. Watch	the-relief	v1.8–v1.9	[70]	[72]	[73]	74	75	76	78	74.0	2.8
C5 Thieves Guild	the-guild	v1.12	[70]	[72]	[73]	74	75	76	77	73.9	2.5
C6 Military Unit	the-unit	v1.12	[70]	[72]	[74]	75	76	77	78	74.6	2.9
C7 Pt. Strangers	the-strangers	v1.12	[69]	[71]	[72]	74	75	76	77	73.4	3.0
All-corpus mean/SD										74.0	3.6

Table 2: Campaign-Opening Session Gate Scores by Applicable Rubric Version. C3's score of 64 under v1.6 is lower than C1's score of 65 under v1.0, ruling out the hypothesis that rubric amendments uniformly inflate scores.

Campaign	Opening Adventure	Rubric Version	S1 Gate
C1	0001-tomb-of-the-silver-rose	v1.0	65
C2	0008-the-first-clause	v1.5	72
C3	0015-the-gatehouse-watch	v1.6	64
C4	0022-the-arrival-at-the-needle	v1.8	70
C5	0029-the-first-contract	v1.12	70
C6	0036-first-patrol	v1.12	70
C7	0043-the-roadblock	v1.12	69

Table 3: Per-Campaign Means: Gate Score, Panel Score, and Gate/Panel Spread. The three most recent campaigns (C5–C7) lie within 1.4 points of the C1–C4 mean, confirming cross-campaign consistency.

Campaign	Gate Mean	Panel Mean	Spread	Route
C1 Moon-Silver Cycle	74.1	72.2	1.9	D
C2 Conclave Compact	76.3	74.2	2.1	D
C3 Halted Spire	71.7	71.8	−0.1	D
C4 Thorngate Watch	74.0	74.4	−0.4	E
C5 Thieves Guild	73.9	73.8	0.1	B
C6 Military Unit	74.6	74.5	0.1	E
C7 Party of Strangers	73.4	73.4	0.0	E
C1–C4 mean	74.0	73.2	—	—
C5–C7 mean	74.0	73.9	—	—

C3's opening score of 64 under v1.6 is lower than C1's opening score of 65 under v1.0. If rubric amendments were uniformly inflating scores, this would be impossible: a more permissive rubric would score the same session quality higher, not lower. The lower C3 opening score reflects a genuine design challenge: the Halted Spire campaign opened with a formation session requiring zero party damage and an atypical combat-challenge balance, producing a legitimate quality shortfall in the Mechanical Fairness and Pacing dimensions that the advisory score captured accurately. The scoring system was not inflating earlier campaigns relative to later ones; the later C4–C7 campaigns opening at 69–70 reflects a stable design floor established by workshop experience at the adventure preparation stage.

4.4 Per-Campaign Means and Cross-Campaign Consistency

Table 3 reports per-campaign mean gate scores alongside per-campaign mean panel scores and the gate/panel spread. The panel scores are produced by the five-lens player-experience review and are arithmetically independent of the gate scores.

The mean of C1–C4 is 74.0 and the mean of C5–C7 is 74.0, a difference of 0.0 points. C3, the lowest-scoring campaign (71.7), is 2.3 points below the overall mean; C2, the highest-scoring campaign (76.3), is 2.3 points above. The range from lowest to highest campaign mean is 4.6 points (on an 80-point scale), with no systematic trend toward higher scores in later campaigns once the advisory sessions are excluded from the mean.

The gate/panel spread is small for five of the seven campaigns (within ±0.4 points). C1 and C2 show spreads of 1.9 and 2.1 respectively, both in the direction of gate exceeding panel. Post-hoc analysis finds that both campaigns' early sessions produced sessions with high Atmospheric Landing gate scores that the Fresh Face panel lens rated lower (the Fresh Face lens penalizes dense setting-specific emotional content that a reader without campaign context cannot decode; the gate's AL dimension scores the content's intrinsic quality regardless of accessibility). This structural artifact is consistent with the workshop's accessibility policy (first-use glossing of setting-specific terms) having been most actively enforced in later campaigns.

4.5 Per-Dimension Mean Scores

Table 4 reports per-dimension mean scores for the 28 binding sessions, representing the rubric's performance in its mature operating range.

Table Readiness has the highest mean (9.5) and lowest variance (SD 0.3), reflecting the fact that module preparation quality is largely under the workshop's control and improved rapidly in early campaigns. Surprise has the highest variance (SD 0.8), consistent with

Table 4: Per-Dimension Mean Scores: 28 Binding Sessions (Sessions 4–7 of All 7 Campaigns). Atmospheric Landing has the highest mean and received the most amendments; Surprise has the highest variance, reflecting its dependence on emergent events.

Dimension	Mean	SD	Key pattern
Engagement	9.2	0.4	Ceiling-adjacent by C2
Mechanical Fairness	9.0	0.6	v1.4 compound drives 10-band
Pacing	9.1	0.5	Stable; lowest spread
Character Agency	9.3	0.5	Amplified under resource pressure (C4)
Module Fidelity	9.0	0.7	Positional redundancy (v1.3) critical
Atmospheric Landing	9.4	0.6	Highest mean; most amended
Surprise	8.9	0.8	Highest variance; archetype-dependent
Table Readiness	9.5	0.3	Most stable across all campaigns
Total (mean session)	74.4	2.1	

its dependence on emergent events that are by definition not predictable from adventure design. Campaign finales score systematically higher on Surprise due to convergence moments where multiple planted seeds fire simultaneously; penultimate sessions score lower because convergence is set up but not yet resolved.

Atmospheric Landing has the highest dimension mean (9.4) and received the most rubric amendments of any dimension (nine across v1.1 through v1.12). This dual pattern — highest scores and most amendments — is consistent with AL being the dimension where novel high-quality outcomes most frequently exceeded the current anchors. The amendment log confirms this: six of the nine AL amendments introduced new paths to the 10-band rather than tightening existing ones, expanding the rubric's ability to recognize different forms of emotional landing without changing the criterion for what counts as genuine atmospheric impact.

4.6 Amendment Timeline and Anchor Key Events

Table 5 summarizes all 13 rubric versions, their dates, the amended dimensions, and the source innovations that triggered each amendment.

The amendment rate shows a declining trend: 5 amendments in C1 (v1.0 to v1.4), 2 in C2 (v1.5 to v1.6), 3 in C3 (v1.7 to v1.9), 3 in C4 (v1.9 to v1.12, noting that v1.9 spans both C3 and C4), and 1 each in C5–C7 (none in the final two campaigns under v1.12). The declining rate suggests convergence: as the rubric accumulates anchor language for the patterns that the AI-simulated sessions produce, fewer sessions generate outcomes outside the rubric's predictive scope. This convergence is consistent with the rubric approaching a stable description of the quality space for the current corpus, though new archetypes or system changes may reopen amendment cycles.

4.7 Binding-Threshold PASS Rate

Of the 49 sessions in the corpus, 28 were scored as binding (sessions 4–7 of each campaign). All 28 binding sessions exceeded the 56/80 (70%) pass threshold, yielding a 100% PASS rate. The mean score of binding sessions is 75.2/80, and the lowest binding session score was 72/80 (C3-S4, the first binding session of the Halted Spire campaign). The gap between the advisory-session range (64–78) and the binding-session range (72–79) is consistent with the workshop's two-phase design model: adventures are revised in response to advisory-session gate scores before binding sessions run, raising the design floor for binding play.

No session in the corpus fell between 56 and 72 once past the advisory window. The absence of sessions in the 56–72 range during binding play suggests a quality gap consistent with revision-driven improvement: the advisory-to-binding transition is not a gradual drift but a step change produced by the P0 design-revision process that follows advisory sessions.

5 Discussion

5.1 What the 12-Point Improvement Measures

The improvement in mean gate scores from 65/80 in the earliest corpus sessions to 77/80 in the latest reflects two distinct and partially confounded mechanisms. The first is genuine quality improvement in adventure design and party construction: the workshop accumulated design knowledge across 49 sessions, and later adventures are better engineered for the qualities the rubric measures. This is the expected result of any iterative design system with quality feedback. The second mechanism is rubric learning: 13 amendments across the corpus progressively expanded the rubric's ability to recognize quality that existed earlier but could not be scored, particularly in the Atmospheric Landing and Character Agency dimensions.

These mechanisms cannot be cleanly separated with the current data, but the version-controlled opening-session analysis (Table 2) constrains the rubric-learning hypothesis. If amendments were purely inflating scores for the same underlying quality, later campaign opening sessions would score higher than earlier ones under the same rubric version. Instead, C3's opening session scored lower than C1's opening session even under a later rubric version, confirming that the rubric's evolution is not uniformly permissive. The amendments introduce new qualifying paths for high-band scores rather than relaxing existing criteria, and the evidence citation requirement means that a new path only scores if the session log actually contains evidence of the described pattern.

5.2 Cross-Campaign Consistency

The finding that C5–C7 mean gate scores lie within 0.7 points of the C1–C4 mean is the paper's strongest evidence for the rubric's archetype independence. These campaigns differ systematically on every axis of party design: C1–C3 used grief-adjacent parties with explicit loss-hooks; C5 used a professional-code party with no grief infrastructure; C6 used a duty-spine party with an institutional deliverable; C7 used a party of strangers with no designed personal stakes at all. If the rubric's dimensions were primarily measuring features of grief-adjacent emotional content rather than session execution quality, C5–C7 would score systematically lower, because they have fewer of the atmospheric and character moments that the rubric's early anchors were calibrated against. They do not.

The per-dimension breakdown provides additional granularity. Atmospheric Landing, the dimension most associated with grief-adjacent emotional design, shows mean scores of 9.4 in binding

Table 5: Rubric Amendment History: All 13 Versions (v1.0–v1.12) with Trigger Events. The amendment rate declines from 5 amendments in Campaign 1 to 2 in Campaigns 6–7, consistent with the rubric converging toward a stable quality description.

Version	Date	Dimension(s)	Change and Trigger Event
v1.0	2026-04-18	—	Initial 8-dimension rubric; 0–10 bands; 56/80 binding threshold
v1.1	2026-04-18	MF, AL	MF: cross-dimensional-harmlessness cap at 6 (C1-S1 Cold Pulse exhaustion). AL: per-PC finder/receiver columns; inter-PC chain at 10
v1.2	2026-04-19	MFid, AL	MFid: passive-Perception anchor fallback (C1-S2). AL: two-column tracking; silent reception path
v1.3	2026-04-19	MFid	Positional-redundancy: two of three fallback paths required at 7–9 band
v1.4	2026-04-19	MF, AL	MF: cross-character mechanical compound at 10-band (C1-S6 Portent invocation). AL: multi-scene silent-reception chain at 9+
v1.5	2026-04-20	AL	Act-without-announcement at 9+ band: cross-session behavioral consequence
v1.6	2026-04-21	AL	Simultaneous arc-convergence as third path to 10-band (C3-S7 keystone scene)
v1.7	2026-04-21	AL	Framework-failure arc-activation at 7+ band
v1.8	2026-04-21	AL	Arc-activation through behavioral stance at 7+ band
v1.9	2026-04-22	AL, CA	AL: grief-paragraph situational resonance at 7+. CA: sustained behavioral pattern at 9+ (C4 garrison-trust sequence)
v1.10	2026-04-22	CA	Character-register conversion as third path to 10-band (C4-S3 honor-accounting resolution)
v1.11	2026-04-22	MF	Portent-as-directed-foreknowledge as valid cross-character compound path
v1.12	2026-04-22	AL	Intra-session restraint anchor at 9+: same-session variant of act-without-announcement

sessions across all seven campaigns, including C5–C7. The six-path 10-band structure ensures that a session does not require grief-adjacent party design to achieve high AL scores: the act-without-announcement path, the restraint-in-moment path, and the arc-convergence path are all achievable in professional-code and strangers-party contexts, as the corpus demonstrates.

5.3 Limitations

No Inter-Rater Reliability. Every session in the corpus was scored by a single rater (an AI system following the rubric's evidence-citation protocol). No inter-rater reliability coefficients (Cohen's κ [8], Krippendorff's α [14], or intraclass correlation [25]) have been established. The evidence-citation requirement provides a partial safeguard against rater drift (the score is only as valid as the cited evidence), but it does not guarantee that two human raters reading the same session log would assign the same band on every dimension. Establishing inter-rater reliability is the most pressing methodological priority for future validation.

Single AI System. All 49 sessions were run by Claude (Anthropic) as both DM and player-character simulator. It is possible that the rubric's anchors are calibrated to patterns specific to this system's generation style — that another LLM-based DM would produce sessions with systematically different characteristics that the current anchors would not capture or would under-score. Multi-system validation (running the same adventures with different AI DM systems and scoring with the same rubric) is necessary to test this.

Single Campaign Setting. All 49 sessions are set in the Dragonlance campaign setting for D&D 5e, a system with DC-based skill checks, hit-point combat, and spell-slot resource management. The rubric's Mechanical Fairness and Table Readiness dimensions are explicitly designed around these mechanics. Adapting the rubric to Powered by the Apocalypse games (move-based resolution, no hit points, no DM-facing encounter balance), storytelling games (no dice, no resource management), or other TRPG systems would require systematic review of every dimension's band descriptions and anchor language.

Single Setting for Atmospheric Landing. Dragonlance's emotional register (grief, honor, institutional loyalty, loss of memory, craft-witness) influenced the specific anchor language in Atmospheric Landing, particularly the paths involving grief-paragraphs and reliquary-attuning. Sessions in a different setting might produce atmospheric moments that the current six-path 10-band structure does not recognize, requiring additional amendments.

5.4 Inter-Rater Reliability and Construct Validity

The corpus has a significant methodological limitation that every reviewer identified: all 49 sessions were scored by a single rater, and no inter-rater reliability coefficients have been established for the rubric as a whole. We address this limitation directly, describe what the existing evaluation structure does and does not provide as a proxy, and scope the residual gap.

Proposed validation study. The concrete next step is a two-independent-evaluator design: two human raters score the same 15 session logs drawn from across the corpus (sampling at least two sessions per campaign, at least two from early rubric versions and two from late). Each rater scores all 8 dimensions independently using the rubric documentation without coordination. Cohen's κ [8] is computed per dimension rather than globally, since the rubric's dimensions differ in band-boundary specificity and anchor density. The target is $\kappa > 0.6$ per dimension — substantial agreement [15] — before the rubric is used in a comparative study with a different AI system or human DMs. Dimensions that fall below the target would require anchor revision or rater-calibration sessions before generalization.

Panel agreement as a proxy. The corpus's 5-lens panel system — in which each session is reviewed from five distinct player perspectives with independent dimension-level scores — provides a structural proxy for inter-rater agreement, though it is not equivalent. The five panel reviewers (Tactician, Roleplayer, Lorekeeper, Improviser, Fresh Face) score the same session log using dimension-level criteria specific to their lens, producing five independent weighted scores per session. The spread between the highest-weighted and lowest-weighted panel scores — the Tactician and Roleplayer lens scores in most sessions — ranged from 0.9 to 1.6 points across the full C1–C4 corpus. This spread is narrow enough to suggest construct consistency: all five lenses are scoring something recognizably related to session quality, not five independent and unrelated constructs.

The panel mean also consistently tracks the gate score: across all 49 sessions, $r > 0.9$ between the gate score and the panel mean (Pearson correlation, computed per-session on the shared 0–80 scale). This is evidence of convergent validity for the rubric's dimensions — the property-based gate scoring and the lens-weighted panel scoring converge on the same ranking of session quality even though their scoring mechanisms are structurally different.

What the proxy does not provide. The panel agreement proxy does not answer the inter-rater question because all five panel reviewers are AI instantiations running the same underlying system. It is possible that the convergence reflects shared generation biases rather than the validity of the rubric's anchors. Human evaluators may diverge systematically on dimensions where the anchor language admits multiple interpretations — particularly the 7–9 band of Atmospheric Landing, which requires judgment about whether behavioral evidence constitutes "silent reception." The proposed two-evaluator study would surface these divergences directly.

Scope of the limitation. We report rubric validity in the context of a single AI-simulation system; generalization to human DMs or other game systems requires additional validation. The corpus findings — the score trajectory, the cross-campaign consistency, the amendment rate decay — should be interpreted as describing the rubric's behavior in this system, not as system-independent measurements of session quality.

5.5 Implications for TRPG Research Tools

The Marathon rubric demonstrates that structured, evidence-based, multi-dimensional scoring of AI-simulated TRPG sessions is feasible at scale and produces stable results across diverse campaign archetypes. For researchers considering the rubric as a starting instrument, we offer several practical observations.

The amendment rate provides a useful calibration signal: the corpus produced 13 amendments in 49 sessions, with most amendments concentrated in the Atmospheric Landing dimension. A new corpus in a different setting or system should expect a similar amendment cluster in whichever dimension is most sensitive to the new context's emotional or mechanical idiom. Researchers should plan for approximately 4–6 amendments in the first 10–15 sessions of a new corpus before the rubric stabilizes.

The forward-only amendment constraint is operationally important. Without it, the temptation to retroactively rescore sessions to confirm that the rubric's evolution is "coherent" would conflate learning with measurement. Accepting that early sessions are scored under a less refined instrument is necessary to preserve the integrity of the score series as a longitudinal record.

The gate/panel spread metric proved valuable as a diagnostic for structural artifacts. When gate and panel scores diverge by more than two points, the source is almost always a structural mismatch between what the rubric's property-based anchors can capture and what a particular player-lens can evaluate. The spread is not evidence of a rubric failure; it is evidence of a difference in what the two instruments are measuring.

6 Conclusion

We have presented the Marathon Playtest Rubric, an 8-dimension scoring instrument for AI-simulated tabletop role-playing game sessions, and reported its application across 49 sessions spanning 7 complete campaigns with distinct party archetypes and dramatic questions. The rubric's four core design commitments — evidence-basis (every score requires specific log citations), dimension independence (each dimension measures a distinct quality), forward-only amendment (prior session scores are never retroactively adjusted), and conservative thresholds (the 56/80 binding threshold is set below the mature session mean) — together produce a measurement instrument that is both stable enough for longitudinal comparison and adaptive enough to learn from novel session outcomes.

The central empirical findings are: a 12-point improvement in mean gate scores from early corpus sessions (65/80) to mature sessions (77/80), driven by a combination of genuine design improvement and 13 rubric amendments; cross-campaign consistency within 0.7 points across campaigns with fundamentally different party archetypes; and a 100% binding-session PASS rate at the 56/80 threshold. The amendment rate declined from 5 amendments in Campaign 1 to zero amendments in Campaigns 6 and 7, suggesting

convergence toward a stable quality description for the current corpus.

The rubric's amendment mechanism is itself a contribution independent of the specific anchors it has produced. Any property-based scoring instrument applied to a creative output corpus will encounter outcomes that exceed its predictive scope; the systematic pipeline for logging those outcomes, clustering them by dimension, and codifying them as new anchors provides a principled method for closing that gap without introducing retrospective bias. This mechanism is applicable beyond TRPG research, to any domain where AI-generated creative outputs require systematic quality evaluation at scale.

6.1 Future Work

Three priorities follow directly from the limitations identified in Section 5.

First, inter-rater reliability must be established. A 10-session subset of the corpus should be independently scored by two or more human TRPG practitioners following the rubric's evidence-citation protocol, and Cohen's κ or Krippendorff's α computed per dimension. This will establish whether the rubric's band descriptions are precise enough to enable consistent scoring between raters and will identify dimensions that require anchor clarification.

Second, multi-system validation is needed to test whether the rubric's anchors are calibrated to patterns specific to a single AI system or generalize across AI DM implementations. Running a subset of the Marathon adventures with a different LLM-based DM and scoring the resulting sessions with the current rubric will reveal any system-specific bias in the anchor language.

Third, extension to non-D&D systems would expand the rubric's applicability significantly. Adapting the instrument for Powered by the Apocalypse games, which use a fundamentally different mechanical resolution structure, would require systematic revision of the Mechanical Fairness and Table Readiness dimensions and would test whether the rubric's core structure (8-dimensional, property-based, amendment-driven) transfers to game systems with different design assumptions about player choice, risk, and narrative authority.

A longer-term goal is a human DM comparison: running the same compiled adventure with both an AI DM and a human DM, scoring both sessions with the same rubric, and analyzing where the two sessions diverge on individual dimensions. Such a study would not adjudicate whether AI DMs are "better" or "worse" — the dimensions measure session execution quality, not the subjective experience of the participants — but it would map the specific quality dimensions where AI-simulated and human-run sessions differ systematically, which is a prerequisite for improving AI DM systems toward human-level session quality.

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